

**AMENDMENT NO. 1 JULY 1994
TO
IS 10634 : 1986 SPECIFICATION FOR BAKERY
SHORTENING**

(First Revision)

(*Page 4, clause 3.1, line 1*) — Insert 'of white to pale yellow in colour' before 'clear in appearance'.

(*Page 5, clause 3.1, line 5*) — Insert 'rancidity or staleness' before 'foreign odour or taste'.

(*Page 5, clause 3.4*) — Substitute the following for the existing clause:

'The material shall contain such quantity of refined sesame oil (*see* IS 547 : 1968), which is sufficient to ensure that when the material is mixed with refined groundnut oil in the proportion of 20 : 80, the red colour produced by the Baudouin test shall not be lighter than 2 red units in 1-cm cell on Lovibond scale when tested as prescribed in Appendix A.'

(*Page 5, clause 3.5*) — Substitute the following for the existing clause:

'The material shall be free from non-edible oils, mineral oil, castor oil and animal fats when tested by the methods prescribed in Sections 9, 10, 12, 14, 15 and 18 of IS 548 (Part 2) : 1976.'

NOTE — Section 18 of IS 548 (Part 2) : 1976 may be used for detecting the presence of animal fat in bakery shortening. Section 21 of IS 548 (Part 2) : 1988 shall be used for confirmatory testing.'

[*Page 6, Table 1, Sl No. (iii)*] — Substitute the following for the existing matter:

'iii) Unsaponifiable matter,	2.0	—	8
percent by mass, <i>Max</i> (<i>see</i> Note)			

NOTE — In case of bakery shortening where proportion of rice bran oil is more than 30 percent by mass, the unsaponifiable matter, percent by mass, *Max* shall be 2.5 percent provided the quantity of rice bran oil of such bakery shortening is declared on the label.'

[*Page 6, Table 1, Sl No. (vii)*] — Add the following new item:

(1)	(2)	(3)	(4)
viii)	Nickel, ppm, <i>Max</i>	1.5	E

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[*Page 7, clause 5.1(h)*] — Renumber the existing as 5.1 (j) and include the following as 5.1 (h):

“Every package of bakery shortening made from more than 30 percent rice bran oil shall bear the following label:

This package of bakery shortening is made from more than 30 percent rice bran oil by mass.”

(*Page 25, Appendix F*) — Add the following new Appendix:

‘APPENDIX F

[*Table 1, Sl No. (viii) and clause 7.1*]

DETERMINATION OF NICKEL IN BAKERY SHORTENING

F-0 Two methods have been prescribed, the colorimetric method and the atomic absorption spectrometric method. The atomic absorption spectrometric method is the reference method.

F-1 COLORIMETRIC METHOD

F-1.1 Reagents

F-1.1.1 Concentrated Ammonia

F-1.1.2 Sodium Citrate — 10 percent.

F-1.1.3 Dimethyl Glyoxime — one percent in ethyl alcohol.

F-1.1.4 Chloroform

F-1.1.5 Ammonia (0.5 M) — 1 : 3.

F-1.1.6 Ammonia — 1 : 1.

F-1.1.7 Hydrochloric Acid — 0.5 M and 1 N.

F-1.1.8 Sodium Hydroxide — 2 N.

F-1.1.9 Potassium Persulphate — 2 percent.

F-1.1.10 Standard Nickel Solution

Dissolve 0.447 9 g of nickel sulphate (NiSO_4) ($6\text{H}_2\text{O}$) in water or 0.1 g of 99.9 percent pure nickel in 10 ml of nitric acid and when dissolution is complete, boil

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gently to expel oxides of nitrogen and dilute to 1 litre. This solution contains 0.1 mg/ml of nickel and may be further diluted to 0.01 mg/ml, if necessary.

F-1.1.11 Perchloric Acid — 6 percent.

F-1.1.12 Bromine Water

F-1.1.13 Nitric Acid — 1 N.

F-1.2 Procedure

F-1.2.1 Preparation of Sample — Representative sample to give a final solution containing greater than 0.05 but less than 10 µg Ni/ml is weighed into a silica dish and heated to flash point using a Bunsen Burner. When the oil catches fire, Bunsen Burner is removed and the oil is allowed to burn in a hood without exhaust. After about 30 minutes the flame ceases on its own and a very small quantity of residue remains. The residue is heated in low Bunsen flame till smoke ceases, before keeping the sample in the muffle furnace. Ashing is completed in the muffle furnace maintaining the temperature at 450°C. Ashing will be complete within 2 hours. Care is needed to take out the sample from the muffle furnace as the ash is very fluffy and quantity is small.

F-1.2.1.1 Dissolve the ash in 1 N nitric acid by warming on the water bath and filter through ashless Whatman filter paper and make up to suitable volume with 1 N nitric acid.

F-1.2.2 Separation of Nickel by Dimethyl Glyoxime-Chloroform Extraction — To an aliquot containing about 50 to 100 ppm of nickel add 5 ml of 10 percent sodium citrate. Neutralize with concentrated ammonia and add a few drops in excess (pH greater than 7.5). Add 2 ml of dimethyl glyoxime (or more as may be required if such copper or cobalt is present, that is, extra volume of 2 ml for each 10 mg of copper and 5 ml for each 10 mg of cobalt). Extract with three 2 to 3 ml portions of chloroform shaking for 30 seconds each time. Shake the combined chloroform extracts with 0.5 ml of 1:3 (0.5 M) ammonia. Repeat the wash with another portion of ammonia if much copper or cobalt is present. Shake the ammonia washings with 1 or 2 ml of chloroform and add the latter to the chloroform extract. Carry out reagent blank using the same quantity of reagents.

Return the nickel to the ionic state by shaking with two 5-ml portions of 0.5 M hydrochloric acid, transfer this solution to a 100-ml volumetric flask and proceed as per calibration.

F-1.2.3 Colour Development

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F-1.2.3.1 Method using persulphate for oxidation

F-1.2.3.1.1 Calibration — Transfer 1, 2, 4, 6, 8 and 10 ml of standard nickel solution (1 ml - 0.01 mg Ni) to six 100-ml volumetric flasks. Add 5 ml of 1 N hydrochloric acid. Dilute to 50 ml and add 1 ml of 10 percent sodium citrate, 3 ml of 2 percent potassium persulphate solution, 15 ml of 2 N sodium hydroxide solution and finally 1 ml of 1 percent ethanolic dimethyl glyoxime solution. Heat to 60 to 70°C and keep at the temperature for 5 minutes. Cool to room temperature and dilute to volume. Measure the absorbance at 465 nm. Plot the photometric readings of the calibration solution against milligrams of nickel present in 100 ml solution.

F-1.2.3.1.2 Develop the colour for sample as described under calibration.

F-1.2.3.2 Method using bromine for oxidation

F-1.2.3.2.1 Calibration — Transfer 1, 2, 4, 6, 8 and 10 ml of standard nickel solution (1 ml-0.01 mg) to six 100-ml volumetric flasks. Add 2 ml of perchloric acid to each flask. Also prepare a separate reference solution by adding 2 ml of perchloric acid to a 100-ml volumetric flask. Dilute to 50 ml and add successively (swirling the flasks between additions) 10 ml of sodium citrate, 5 ml of bromine water and adjust sufficient ammonia (1:1) to bleach the bromine colour. Add 3 ml of ammonia (1:1) in excess and cool to room temperature. Add 3 ml of dimethyl glyoxime solution, dilute to the mark and mix (the addition of bromine water, ammonia and dimethyl glyoxime should be made without interruption and subsequent photometric measurement should be made after 10 ± 2 minutes after addition of dimethyl glyoxime). Transfer a suitable quantity of the solution to an absorption cell, cool and measure the absorbance at 530 nm. Plot the photometric readings of the calibration solution against milligrams of nickel per 100 g of solution.

F-1.2.3.2.2 Develop the colour for sample as described under calibration and read the absorbance against reagent blank.

F-1.2.4 Calculation

$$\begin{array}{lcl} \text{Concentration of nickel} & & \text{Dilution factor} \\ \text{in sample (ppm)} & = & \frac{\text{Mass of sample}}{\text{in g}} \\ & \text{Measured} & \\ & \text{concentration} & \\ & \text{(from calibration} & \\ & \text{curve)} & \end{array} \times$$

F-2 ATOMIC ABSORPTION SPECTROMETRIC METHOD

F-2.1 Reagents

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F-2.1.1 Stock Nickel Solution — Dissolve 1 g of 99.99 percent nickel in minimum volume of 1 + 1 HNO₃ and dilute to one litre with 1 percent nitric acid.

F-2.1.2 Nitric Acid — 1N.

F-2.2 Procedure

F-2.2.1 Preparation of Sample — Representative sample to give a final solution containing greater than 0.05 but less than 10 µg Ni/ml is weighed into a silica dish and heated to flash point using a Bunsen Burner. When the oil catches fire, Bunsen Burner is removed and the oil is allowed to burn in a hood without exhaust. After about 30 minutes the flame ceases on its own and a very small quantity of residue remains. The residue is heated in low Bunsen flame till smoke ceases, before keeping the sample in the muffle furnace. Ashing is completed in the muffle furnace maintaining the temperature at 450°C. Ashing will be complete within 2 hours. Care is needed to take out the sample from the muffle furnace as the ash is very fluffy and quantity is small.

F-2.2.1.1 Dissolve the ash in 1 N nitric acid by warming on the water bath and filter through ashless Whatman filter paper and make up to suitable volume with 1 N nitric acid.

F-2.2.2 Calibration of Atomic Absorption Spectrometer — Calibrate the instrument with 0.2, 0.4, 0.8, 1.6, 2.0, 4.0, 8.0 and 10 µg/ml standard nickel solution by setting the wavelength of 232 nm using air acetylene flame and 1N nitric acid as blank. The absorbance is plotted against concentrate.

F-2.2.3 Measurement — Run the sample solution into the instrument and calculate the concentration of nickel in the sample.

F-2.2.4 Calculation

$$\begin{array}{lcl} \text{Concentration of nickel} & & \text{Measured} \\ \text{in sample (ppm)} & = & \text{concentration} \times \frac{\text{Dilution factor}}{\text{Mass of sample in g}} \end{array}$$

(FAD 44)